

Lesson 9: Inverse Functions

Criteria	Exceeds Expectations (5)	Meets Expectations (3-4)	Below Expectations (1-2)
Finding Inverse	Correct algebraic steps, isolates perfectly, and uses proper notation $f^{-1}(x)$.	Mostly correct steps with minor errors in algebraic manipulation or notation.	Unable to correctly swap variables or solve for the new .
Verification	Accurately performs composition $(f \circ f^{-1})(x)$ and $(f^{-1} \circ f)(x)$ to prove results are x.	Verifies in one direction but fails the other, or makes a small simplification error.	No verification shown or verification does not lead to x.
Graphing	Accurately graphs $f(x)$ and $f^{-1}(x)$. with correct reflection over $y = x$ and key points plotted.	Graphs both functions with general shape, but reflection or key points are slightly off.	Graphs do not show the inverse relationship or reflection.
Conceptual Understanding	Identifies domain/range restrictions necessary for non-one-to-one functions.	Understands when a function is one-to-one, but struggles with domain/range constraints.	Does not understand the one-to-one requirement.